



ZENITH AERO: GRAND VOYAGER (305cm LWB)

COMMUNITY EMPOWERMENT & ACCESSIBILITY DOSSIER

A SUPERIOR PEDICAB FOR LOCAL RESILIENCE

The **Zenith Aero: Grand Voyager** is a high-performance, accessible vehicle designed to be hand-built using sustainable materials and simple fabrication methods. It is an affordable machine built for the rainy nights of Vancouver—designed to empower local entrepreneurs and provide families with a dry, stable, and superior transport solution.

1. VISION: THE ACCESSIBLE VOYAGER

The Grand Voyager is more than a vehicle; it is a tool for economic independence. By using locally available bamboo and a simplified hybrid core, we create a machine that is superior in comfort but remains hand-built without the need for expensive CNC equipment or hard-to-find components.

2. THE VANCOUVER USE-CASE: FAMILY & CARGO

Designed for the Pacific Northwest, the 305cm Long-Wheelbase (LWB) chassis provides the stability and space required for daily life:

- **SUPERIOR COMFORT:** The artisan monocoque absorbs road vibration better than steel, ensuring children and groceries stay stable and protected.
 - **WEATHER READY:** The modular chassis is designed to support future all-weather enclosures, keeping families dry on rainy Vancouver commutes.
 - **HIGH CAPACITY:** A 250kg payload allows for a full week's groceries plus two passengers with ease.
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3. ENGINEERING: HAND-BUILT SUPERIORITY

The Grand Voyager utilizes the "**Blue Board**" **Molding Method**—an artisan technique that uses high-density foam jigs to achieve complex, organic curves.

TECHNICAL SPECIFICATIONS (COMMUNITY EDITION):

- **CHASSIS:** 14-layer Bamboo with a locally sourced Carbon Fiber spine.
- **FABRICATION:** Hand-laminated using slow-cure bio-epoxy (Swell Composites).
- **MAINTENANCE:** Designed for easy repair using common workshop tools (sanders, hand-grinders).
- **PROPORTIONS:** 3050mm total length for maximum stability.

4. ECONOMIC IMPACT: EMPOWERING THE LOCAL WORKFORCE

The Grand Voyager is structured to minimize initial Capital Expenditure (CAPEX), making it an ideal business-in-a-box for those looking to start their own mobility service.

- **HAND-BUILT SCALE:** No high-overhead factory is required. Small hubs can produce these vehicles using local materials.
- **REDUCED BARRIERS:** By utilizing renewable bamboo (pennies per pound), the cost of the raw frame is significantly lower than aluminum or steel alternatives.
- **MODULAR GROWTH:** Future accessory phases (like cargo racks or shelters) can be added as the user's business grows.

5. LOCAL SUPPLY CHAIN (VANCOUVER, BC)

We prioritize partners that support a sustainable, local circular economy:

- **Coolite Bamboo:** Heat-treated poles for high-tensile strength.
- **Swell Composites:** Eco-friendly bio-resins for structural bonding.
- **Industrial Plastics:** Local sourcing for reinforcement materials.

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